

Technical Architecture

Secure, evidence-aware digital twin architecture for auditable energy analysis and investment decisions.

Team EnerGen AI · Project Irene
T1 — AI for Clean Energy



Project terms used throughout this document

TERM	DEFINITION	TERM	DEFINITION
Irene	Governed energy intelligence: deterministic analysis with optional model-assisted language.	Energy digital twin	Traceable link across assets, meters, operations and evidence.
Deterministic analysis	Rules, formulas and optimisation return the same result from the same confirmed inputs.	Local-first	Core analysis runs without sending raw files to a cloud model.
Optional cloud enhancement	Consent-based server support for language and selected document recognition.	Evidence class	Measured, documented, assumed, modelled or synthetic.
Human confirmation gate	Reviewer approval before mapped fields or extracted facts enter a client model.	Canonical data contract	Standard units, time, meter hierarchy and source identity.

Abbreviations: EUI = energy use intensity · M&V = measurement and verification · BESS = battery energy storage system · SOC = state of charge · NMAE = normalized MAE.

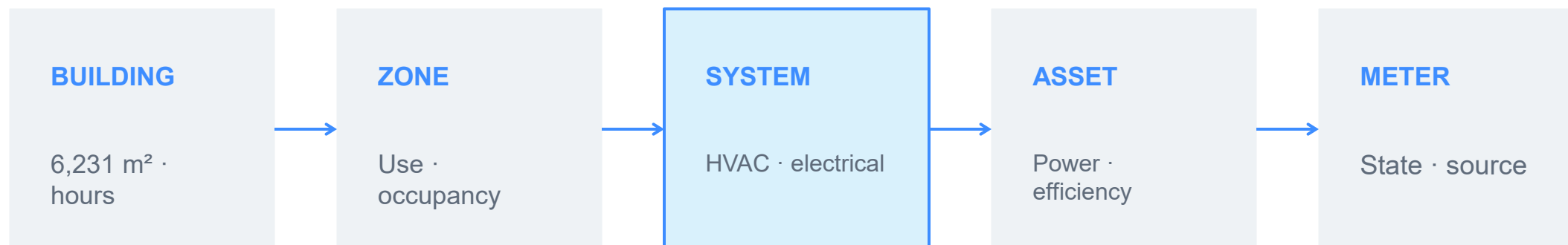
Irene connects through governed adapters



Migration: inventory → map → backfill → reconcile → shadow run → approved cutover. API/SQL/SFTP/gateway connectors are configured per client.

Evidence links every asset and value

Entities link physical scope, operating state, engineering constraints and the evidence behind every value.



The anonymized reference case links approved aggregates, synthetic public records and quality evidence to a building-system-meter graph and 106.14 kWp of PV.

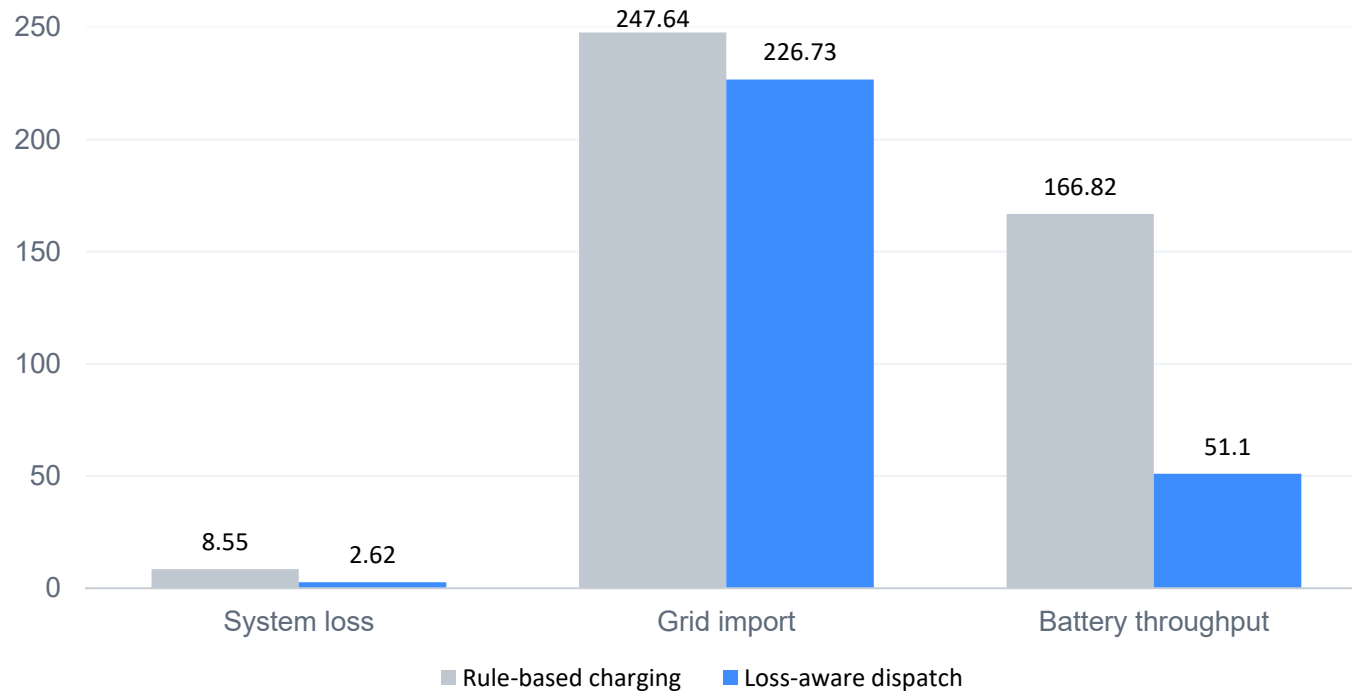
Evidence quality governs every model layer

MODEL LAYER	AVAILABLE EVIDENCE	DATA REQUIREMENT	VALIDATION CHECK
Energy baseline	12 approved monthly anchors	15-min meter + schedules	Month sums + residuals
Forecasting	Calendar prior · 14.36% NMAE	Interval load + indoor sensors	Time split · MAE · RMSE
Digital twin	Reference graph + confirmed client project	Zone/circuit/equipment map	Source trace + mapping + constraints
Optimisation	Storage dispatch optimisation	BESS design + supplier quote	Cost · loss · SOC · violations

Evidence boundary: 14.36% NMAE validates a synthetic reference time-shape only; site-hourly load remains estimated until authorized interval data are available.

Loss-aware optimisation reduces storage losses

Design study only: the reference case has no installed battery. Both strategies use identical constraints and terminal state of charge.



69.4%

lower conversion loss

69.4%

less battery throughput

8.4%

less grid import

One governed stack supports audit, pilot and portfolio scale



Target customers: commercial buildings · campuses · industrial parks · energy service companies (ESCOs). Local-first by default; client files stay governed.

Measured first. Optimised second.

Validated on a Ningbo reference case; designed for configurable deployment in Malaysia, pending local pilot validation.

IMPLEMENTATION PATH

File audit → temporary metering → calibration → savings verification → multi-site scale.